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SHARPCUT

FLATBED CUTTER

Maintenance Manual

COLEX SHARPCUT DAILY MAINTENANCE:

- Clean any dust and debris that may collect on the table surface, different parts of the Colex cutter and underneath the machine (failure to do so could result in blown vacuum motors).
Note: Machine must be turned “off” before cleaning.
- Check the debris collector to be sure it is not full. Dump the debris when necessary to prevent the clogging of the debris collector hose.
- Adjust the conveyor belt rolls positions if necessary.

COLEX SHARPCUT MONTHLY MAINTENANCE:

- Clean dust and debris that has built up on the rails and racks, if necessary. See images for details.
- Grease 12 fittings with LOW VISCOSITY GREASE. See images for details.
- Oil the 2 fittings on the sides of Stations 2 (fitting is on the right side) and 3 (fitting is on the left side). Place the specific material (vinyl) underneath the tool head to prevent the dripping of the oil on the table surface. After oiling the fittings rotate W motor clockwise and counterclockwise to spread the oil evenly. See images for details.
- Use Teflon Lithium Spray on the racks of the X and Y axis and move the head of the machine left and right and the bridge back and forth to spread evenly. Also, lubricate the Z axis ball screw located behind the tool head. It will be visible if the tool head is moved down. To spread it evenly move the head up and down.

Maintenance products are available to order at colex.com.

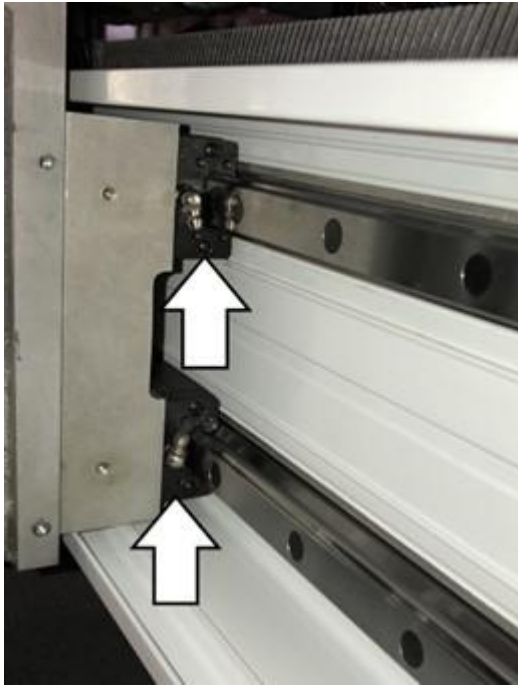
CLEANING THE SHARPCUT

One of the most important parts of maintaining the Colex machine is **keeping it clean**. It will extend the life of the machine and the quality of the jobs done by it. Make sure to always keep the table surface clean.

It's important to keep the **area around the machine** clean, especially if you use a conveyor belt because if we are advancing the conveyor the debris underneath of the machine will move onto the surface of the table.

It is very important to make sure to **clean the Measuring Device** on the Colex SharpCut. If you do not clean the device, it will affect the accuracy of all your cuts on the machine. If there is too much debris underneath the plate of the measuring device it can be extremely dangerous, so we advise always making sure it is clean.

Cleaning the Limit Sensors is an important part of keeping moving parts of the machine properly adjusted and positioned. Debris and grease on the sensors might affect the machine's operations and the quality of the finished products.



GREASING THE “Y” AND “X” AXIS

Grease a total of 4 fittings on the “Y” axis. Two are located on each side of the tool head. (See image on the left).

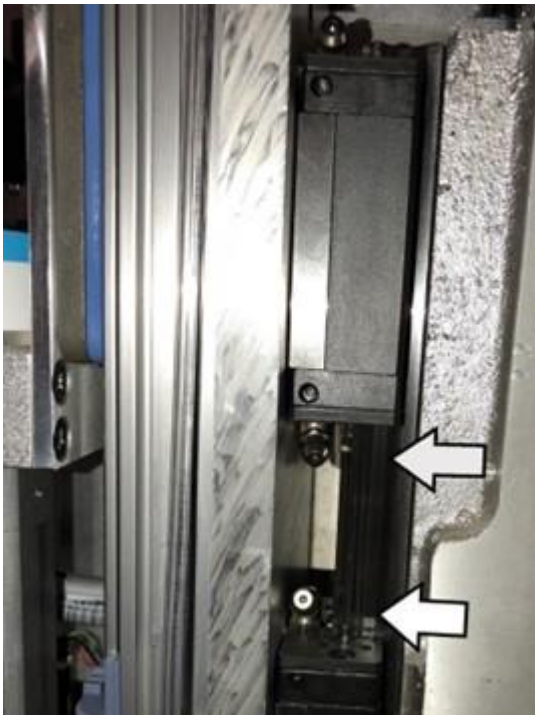
Grease a total of 4 fittings on “X-1” and “X-2” axis on both sides of the bridge. (See image on the left).

Please pump the grease gun no more than two times. (Pumping more than necessary can damage the bearings).

PLEASE USE THE LOW VISCOSITY GREASE (Part # T00855) MONTHLY.

Maintenance can vary depending on how often the machine is used

GREASING THE “Z” AXIS



Grease a total of 4 fittings on the “Z” axis. Two are located behind each side of the tool head. (See images)

Please pump the grease gun no more than two times. (Pumping more than necessary can damage the bearings).

PLEASE USE THE LOW VISCOSITY GREASE (Part # T00855) MONTHLY.

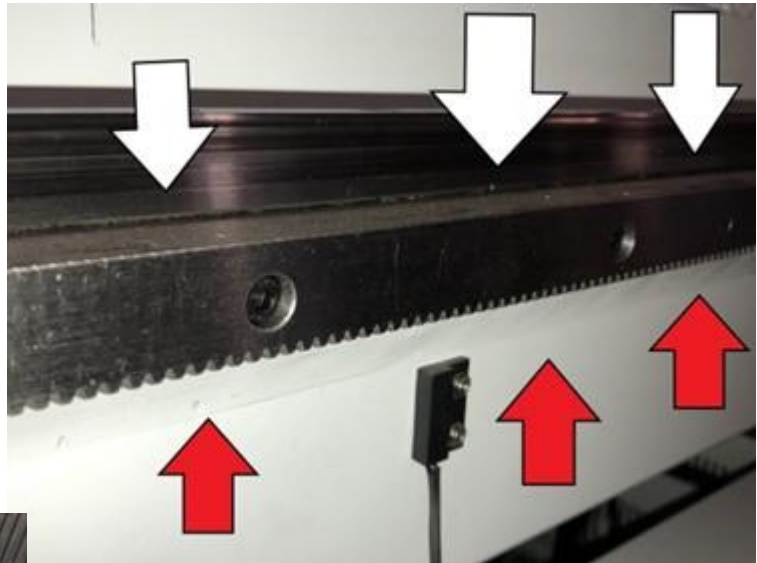
Maintenance can vary depending on how often the machine is used

MAINTAINING THE RAILS AND RACK AND PINION ASSEMBLY

Clean off debris that has built up the rails when it is necessary. (White arrows on the image)

MONTHLY use **WHITE LITHIUM SPRAY** (Part # T00857) on the rack and pinion assembly of the X and Y axis. (Red arrows on the image)

Move Bridge and Tool Head back and forth for even coverage.



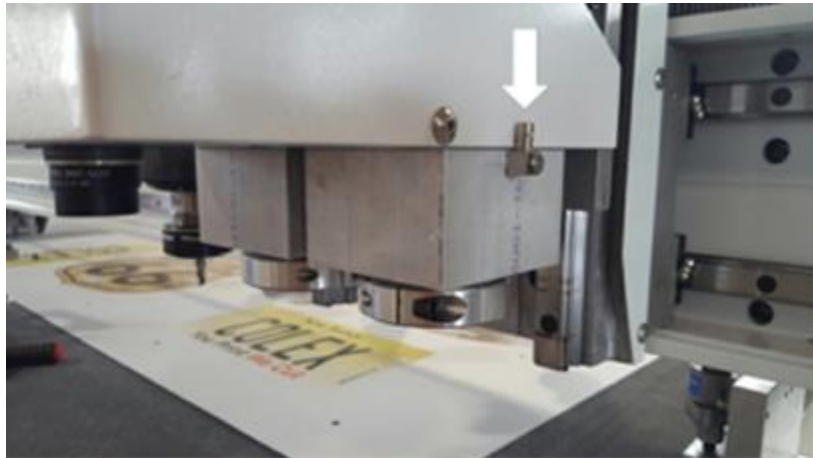
Avoiding buildup of debris: To avoid build up, clean off debris that has built up on the rails on Y, X1, X2 and Z axis when necessary. (See image).

***Maintenance can vary depending on how often the machine is used



OILING FITTINGS OF THE STATIONS #2 AND #3

Oil the two fittings beside Tool #2 and Tool #3 with ZOOM SPOUT OILER (Part # T0P128) **MONTHLY**.
(See images)



DAILY MAINTENANCE WHILE ROUTING

Clean the table surface when necessary.

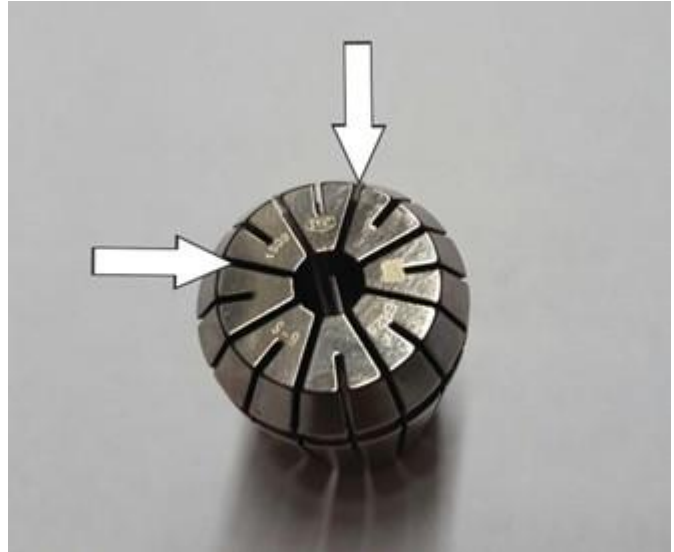
FAILURE TO CLEAN THE TABLE SURFACE MAY RESULT IN THE WEAKENING OF VACUUM TABLE SYSTEM AND THE ACCURACY OF THE CUTS!

If you are routing daily and frequently you **MUST** check for debris built up in between the slots of the collet. Debris that is built up can stop the collet from clamping down securely on the router bit. (See image).

Please clean the collet every time you replace the router bits!

FAILURE TO CLEAN THE COLLET CAN RESULT IN DAMAGE TO THE TABLE/BIT!

Make sure the router collet and the nut are clean. If you route a lot, make sure to clean it every day. If you run all day with the router clean it twice per day.



Make sure that when you tighten the collet/router bit, you use the proper wrenches for the collet being used.

Dirty collets and router bits not being tight enough will result in table surface damage. This is one of the possible scenarios of what happens to the conveyor belt and the acrylic/polycarbonate sheet underneath it. The marks on the top of the sheet indicate the gradual lowering of the bit into it during the cut.

Check your debris collector daily if you are routing frequently. Lift up on the handles located on the sides to empty. (See image)

***Maintenance can vary depending on how often the machine is used

Daily Maintenance while Using the Conveyor

Conveyor Belt Alignment Procedure

Step 1: Move the top two rollers as close to the machine as possible without the roller touching the machine or the bracket mounting screws.

Step 2: Adjust the bottom two rollers so that they are parallel to the top roller. To do this measure from the top roller support to the bottom roller support and adjust the bottom roller up/down. The goal is to get the bottom roller as close to parallel to the top roller as possible. The belt should be loose enough to shift the belt side to side by pulling on the unsupported web of the belt.

Step 3: At this time advance the conveyor belt several times and see which way the belt is tracking. Standing at the feed/back side of the cutter observe if the belt is tracking to the left or the right.

Step 4: If Tracking Left: Shift the right side of the top feed/back roller away from the machine.

If Tracking Right Shift the left side of the top feed/back roller away from the machine.

Step 5: Repeat Process 3 and 4 until the belt no longer rubs against the side clamps.

Note: The belt should be loose enough to shift side to side by hand at all times. If this cannot be achieved move the bottom rollers up to loosen the belt.



Weekly Maintenance while Using Knives, Drawing Tool, Scoring or Perforating Wheels

Lubricate all four sides of the tools installed into stations 2 and 3 with the oil provided in your Colex maintenance kit to prevent the tools from getting stuck in the stations. Use 1-2 drops of oil on a clean rag or paper towel.

If there are scratches on the sides of the tools, please remove them using fine sandpaper or scotch-brite. Please oil the sides of the tools after removing the scratches.

Air pressure settings are an important part of Colex SharpCut operations. The PSI can be controlled on the air regulator at the back of the unit.

PSI settings should be consistent. It should never be lower than 65 psi.

Air sensors installed on Colex units can trigger the error message "Check Air Pressure (min 55 psi)" and pause the machine's operations until resolved. To avoid this and any other possible issues related to the main air pressure - PLEASE KEEP the PSI level of the air pressure at least at 65 PSI (4.5 Bar).

CAUTION: Low air pressure can cause a depth of cut inconsistency. Having this issue in the air line can result in major cutting surface damage and as a result - delayed projects, lost revenue and possibly clients, and the need for service by certified Colex technicians or engineers.

An important part of air pressure maintenance is compressor maintenance. Please follow all the maintenance procedures required by your compressor manufacturer.

CAUTION! One of the maintenance operations which is often forgotten when using the compressor: it's important PERIODICALLY to release the water from the compressor tank by opening its valve. Most compressors produce condensation inside of the tanks. That is why it's important to follow the recommendations and instructions provided in the compressor manuals by its manufacturer.

To access more detailed maintenance procedures and their description please watch the videos and read the PDF files available in Colex Support Center.

Unit Conversion Table

CONVERSION TABLE

Fractions to Decimals to Millimeters

Fractions	Decimals	Millimeters	Millimeters	Decimals	Fractions
$\frac{1}{64}$	.0156	.3969	13.0969	.5156	$\frac{33}{64}$
$\frac{1}{32}$	.0312	.7938	13.4938	.5312	$\frac{17}{32}$
$\frac{3}{64}$	.0469	1.1906	13.8906	.5469	$\frac{35}{64}$
$\frac{1}{16}$	.0625	1.5875	14.2875	.5625	$\frac{9}{16}$
$\frac{5}{64}$	.0781	1.9844	14.6844	.5781	$\frac{37}{64}$
$\frac{3}{32}$	.0938	2.3813	15.0813	.5938	$\frac{19}{32}$
$\frac{7}{64}$	.1094	2.7781	15.4781	.6094	$\frac{39}{64}$
$\frac{1}{8}$	.125	3.1750	15.8750	.625	$\frac{5}{8}$
$\frac{9}{64}$	.1406	3.5719	16.2719	.6406	$\frac{41}{64}$
$\frac{5}{32}$	.1562	3.9688	16.6688	.6562	$\frac{21}{32}$
$\frac{11}{64}$	.1719	4.3656	17.0656	.6719	$\frac{43}{64}$
$\frac{3}{16}$	.1875	4.7625	17.4625	.6875	$\frac{11}{16}$
$\frac{13}{64}$	.2031	5.1594	17.8594	.7031	$\frac{45}{64}$
$\frac{7}{32}$	.2188	5.5563	18.2563	.7188	$\frac{23}{32}$
$\frac{15}{64}$	.2344	5.9531	18.6531	.7344	$\frac{47}{64}$
$\frac{1}{4}$	.250	6.3500	19.0500	.750	$\frac{3}{4}$
$\frac{17}{64}$	.2656	6.7469	19.4469	.7656	$\frac{49}{64}$
$\frac{9}{32}$	.2812	7.1438	19.8438	.7812	$\frac{25}{32}$
$\frac{19}{64}$	.2969	7.5406	20.2406	.7969	$\frac{51}{64}$
$\frac{5}{16}$	.3125	7.9375	20.6375	.8125	$\frac{13}{16}$
$\frac{21}{64}$	.3281	8.3344	21.0344	.8281	$\frac{53}{64}$
$\frac{11}{32}$	.3438	8.7313	21.4313	.8438	$\frac{27}{32}$
$\frac{23}{64}$	.3594	9.1281	21.8281	.8594	$\frac{55}{64}$
$\frac{3}{8}$	.375	9.5250	22.2250	.875	$\frac{7}{8}$
$\frac{25}{64}$	.3906	9.9219	22.6219	.8906	$\frac{57}{64}$
$\frac{13}{32}$	.4062	10.3188	23.0188	.9062	$\frac{29}{32}$
$\frac{27}{64}$	.4219	10.7156	23.4156	.9219	$\frac{59}{64}$
$\frac{7}{16}$	.4375	11.1125	23.8125	.9375	$\frac{15}{16}$
$\frac{29}{64}$	.4531	11.5094	24.2094	.9531	$\frac{61}{64}$
$\frac{15}{32}$	.4688	11.9063	24.6063	.9688	$\frac{31}{32}$
$\frac{31}{64}$	.4844	12.3031	25.0031	.9844	$\frac{63}{64}$
$\frac{1}{2}$	.500	12.7000	25.4000	1.000	1

COLEX SHARPCUT METRIC/INCHES (DECIMALS AND FRACTIONS) CONVERSION CHART

What is one thousandth of the inch (0.001")? It's almost impossible to see it with the human eye. We know that thickness of the paper is about 0.004", so one thousandth of the inch is 4 times less than the thickness of the paper.

Millimeter, MM	INCH, decimal s	INCH, fractions	Millimeter, MM	INCH decimals	INCH fractions
1mm	0.039"	> 3/64"	12mm	0.472"	< 15/32"
1.5mm	0.059"	> 1/16"	12.7mm	0.5"	1/2"
2mm	0.079"	< 5/64"	15mm	0.591"	> 19/32"
3mm	0.118"	> 1/8"	15.87mm	0.625"	3/8"
3.175mm	0.125"	1/8"	19.05mm	0.75"	3/4"
4mm	0.157"	< 5/32"	22.225mm	0.875"	7/8"
5mm	0.197"	> 13/64"	25mm	0.984"	> 1"
6mm	0.236"	< 15/64"	25.4mm	1.000"	1"
6.35mm	0.25"	1/4"	31.75mm	1.25"	1" and 1/4"
8mm	0.315"	< 5/16"	38.1mm	1.5"	1" and 1/2"
9.52mm	0.375"	3/8"	44.45mm	1.75"	1" and 3/4"
10mm	0.394"	< 25/64"	50.8mm	2"	2"

Colex provides this conversion chart to make it easier for machine operators to work with different materials and we hope it will help you to find proper blades and router bits for your specific material thickness.